Skills Summary

Unit Fractions in Area Models

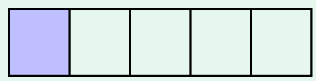
Use this reference while completing your worksheet or when studying.

1. Count the equal parts, then name the shaded one

Bottom number: how many *equal parts* the whole shape is cut into. **Top number:** how many parts are shaded.

When exactly one part is shaded, the top number is 1 and the fraction is called a **unit fraction**: $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{5}$, ... Count the *parts*, never the cut lines — three lines across a bar make four parts.

Example: Name the shaded part.



Step 1. One part is shaded, so this will be a unit fraction — the only thing to work out is the bottom number.
Step 2. Touch and count the equal parts across the bar: one, two, three, four, five.
One of five equal parts: $\mathbf{\frac{1}{5}}$

Try it: A circle is cut into 6 equal slices and one slice is shaded. What fraction is the shaded slice?

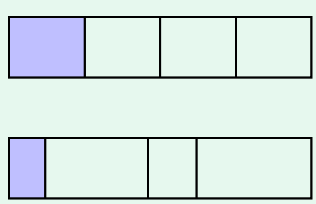
check: $\mathbf{\frac{1}{6}}$

2. First check: are the parts really the same size?

A fraction name only fits when the whole is cut into parts of the *same size*. Pieces that are all different are still pieces of the whole — they just have no fraction name yet.

So look before you count: *equal parts?* then *how many?*

Example: Which bar's shaded piece can be named?



Step 1. Compare the pieces inside each bar first — only the bar with equal pieces can carry a fraction name.
Step 2. The top bar has four pieces all the same width; the bottom bar's four pieces are all different, so it is out.
Top bar, one of four equal parts: $\mathbf{\frac{1}{4}}$

Try it: A square is cut into 3 strips of exactly the same size and one strip is shaded. What fraction is shaded?

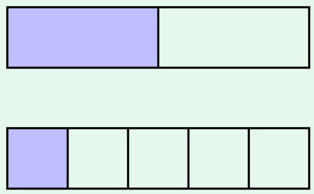
check: $\frac{1}{3}$

3. More cuts make smaller pieces

Two unit fractions of the *same whole*: the one with the bigger bottom number is the *smaller* piece, because the whole was chopped into more pieces.

$>$ means *is greater than* $<$ means *is less than*

Example: Which shaded piece is bigger?



Step 1. Both bars are the same length and both have one part shaded, so only the size of one part decides this.

Step 2. Two cuts of the whole give a piece half the bar; five cuts give a much narrower piece.

The half is bigger: $\frac{1}{2} > \frac{1}{5}$

Try it: Two same-size pizzas: one cut into 6 equal slices, one cut into 3. Which sign makes $\frac{1}{6}$ _____ $\frac{1}{3}$ true?

check: $<$

(!) Watch out: count the *pieces*, not the cut lines — one line down the middle of a bar makes two parts, not one. And a bigger bottom number never means a bigger piece: $\frac{1}{8}$ is smaller than $\frac{1}{2}$, even though 8 is bigger than 2.